A FLAT INJECTIVE EPIMORPHISM OUT OF A RING OF WEAK DIMENSION AT MOST ONE LANDS IN AN INTEGRAL EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted proof span contains a displayed chain of three tensor product maps; it is delivered verbatim. The three references inside the counted proof span (two Algebra references and the weak dimension statement) are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- A ring element integral over a subring generates a finite ring extension. invoked at proof lines 30713-30715. Makes A' = A[x] a finite A-module.
- A finite ring epimorphism is surjective. invoked at proof lines 30716-30717. Reduces A = A' to the epimorphism property of A -> A'.
- Submodules of flat modules are flat when the base has weak dimension at most one. invoked at proof lines 30718-30719. Gives flatness of A' over A, hence injectivity of the tensor square map.

Source reference. Original lemma 02JJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3338–3338. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04VT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6336–6336. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 30575: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30575–30575. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a ring of weak dimension at most 1. If $A \to B$ is a flat, injective, epimorphism of rings, then A is integrally closed in B.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $x \in B$ be integral over A. Let $A' = A[x] \subset B$. Then A' is a finite ring extension of A by Algebra, Lemma 02JJ. To show $A = A'$ it suffices to show $A \to A'$ is an epimorphism by Algebra, Lemma 04VT. Note that A' is flat over A by assumption on A and the fact that B is flat over A (Lemma 30575). Hence the composition

$$A' \otimes_A A' \to B \otimes_A A' \to B \otimes_A B \to B$$

is injective, i.e., $A' \otimes_A A' \cong A'$ and the lemma is proved. □